SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA04 **COURSE CODE:** Operating Systems

COURSE OUTCOME COVERED

CO1: *Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments. (BL2, BL3)*

QUESTIONS: 1 **Total Marks:100**

Annexure A

Concept Mapping

Code of Conduct:

I Srimathi B (192511448) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Srimathi B

Annexure A

Concept Mapping

Question 1
Show a multi-level concept map integrating:
Operating System Types → OS Structures → System Calls → OS Functions.
Your map must show cross-links (e.g., how OS structure influences system call implementation). **(BL2)**

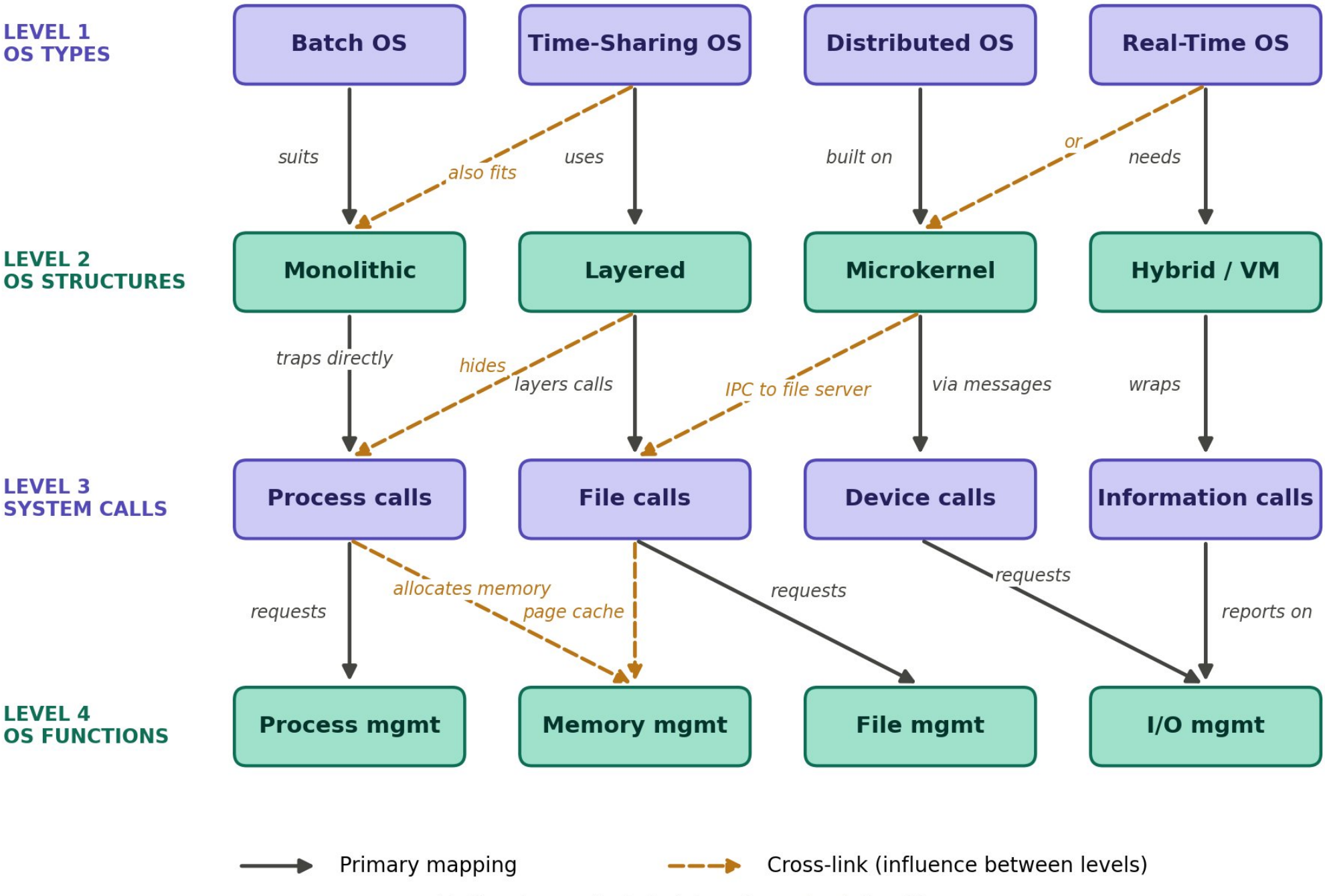
Question 2
Construct a comprehensive concept map representing:
Process Management, Memory Management, File System, and I/O System.
Include interdependencies and indicate where system calls interact with each component. **(BL3)**

Question 3
Sketch a comparative concept map for:
Batch OS, Time-Sharing OS, Distributed OS, and Real-Time OS.
Highlight differences, similarities, advantages, and limitations using linking phrases. **(BL2)**

Question 4
Illustrate a dynamic concept map showing the execution flow:
User Program → API → System Call Interface → Kernel → Hardware → Response back to User.
Include feedback loops and control flow paths. **(BL3)**

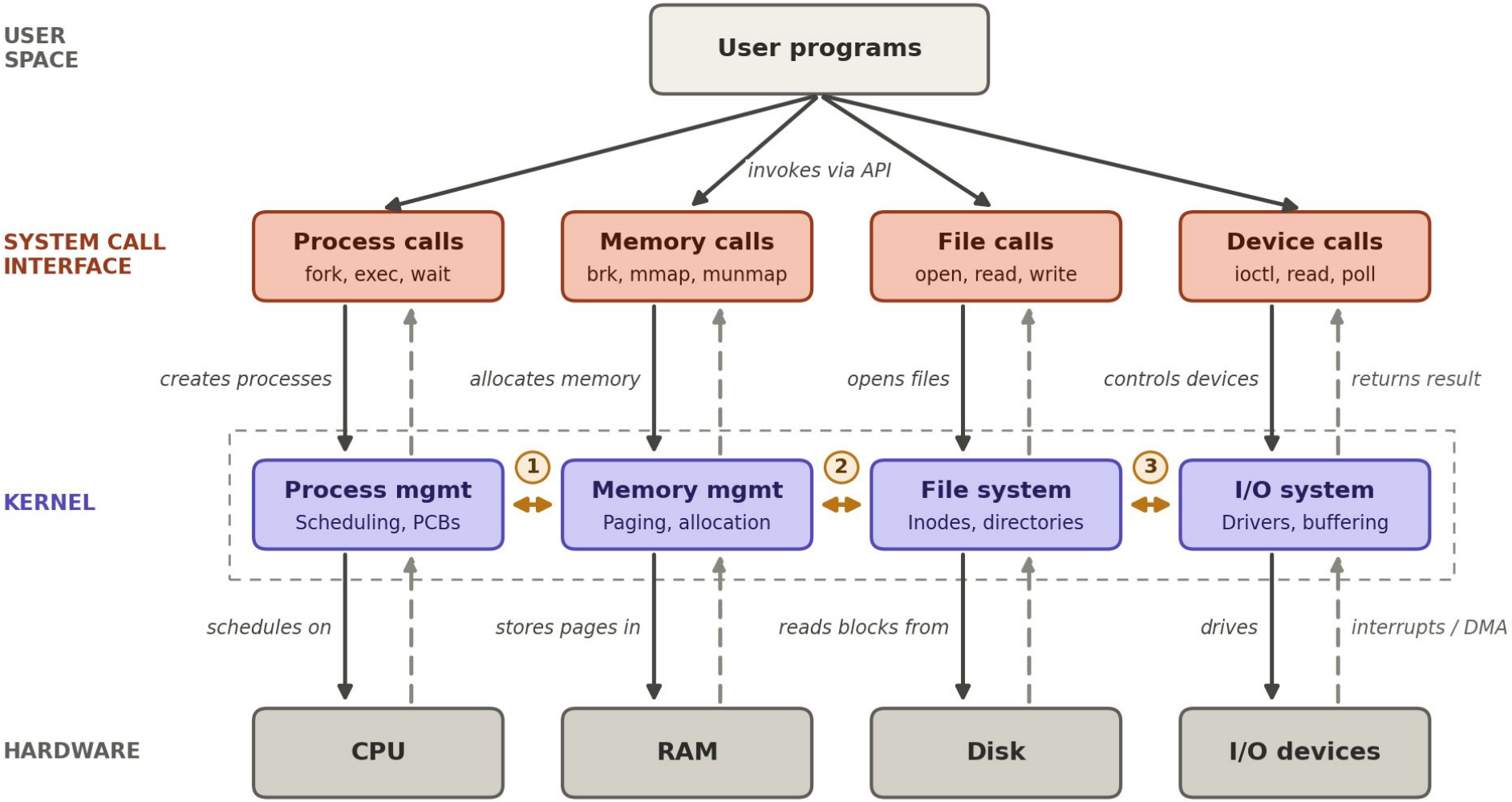
Question 5
Sketch a hierarchical + functional concept map for system calls:
Process, File, Device, and Information system calls.
Also show how they relate to OS services like scheduling, memory allocation, and device handling. **(BL3)**

Concept Map: OS Types → OS Structures → System Calls → OS Functions



Linking phrases (in italics) describe each relationship

Concept Map: Process, Memory, File System and I/O System



Request / control Response / interrupt Interdependency

1 Process ↔ Memory: each process needs its own address space; fork/exec allocate memory

2 Memory ↔ File system: mmap and the page cache map file data into memory

3 File system ↔ I/O: file read/write become block requests handled by device drivers